SEQUENCE 1

Arithmetic

2–11 September 2026

1.1 Integers, multiples and divisors

Key point · Definitions. $\mathbb{N} = \{0, 1, 2, \ldots\}$ is the set of natural numbers. $\mathbb{Z}$ contains the natural numbers and their opposites. For $a, b \in \mathbb{Z}$, saying that a is a **multiple of** b means that there exists $k \in \mathbb{Z}$ such that $a = bk$. We also say that b divides a.

$18 = 6 \times 3$: 18 is a multiple of 3 and of 6. Every integer is a multiple of 1 and of itself; 0 is a multiple of every integer. The only multiple of 0 is 0.

Example 1. List the positive divisors of 18 and of 24. Write three multiples of 7, one of them negative. Is 35 a multiple of 5? of 6? Justify.

Solution. Divisors of 18: $1, 2, 3, 6, 9, 18$. Divisors of 24: $1, 2, 3, 4, 6, 8, 12, 24$. For example $-7, 0, 14$ are multiples of 7. $35 = 5 \times 7$ is a multiple of 5; $35 = 6 \times 5 + 5$ is not a multiple of 6.

Key point · Sum of multiples. The sum of two multiples of an integer a is a multiple of a.

Proof to write.

Solution. Write $u = ak$ and $v = a\ell$, with $k, \ell \in \mathbb{Z}$. Then $u + v = a(k + \ell)$ and $k + \ell \in \mathbb{Z}$. So $u + v$ is a multiple of a. This also holds for $a = 0$.

Fluency · 5–10 min. Calculate $\frac{2}{3} + \frac{1}{6}$; $3(x + 2) - x$; the image of 2 under $x \mapsto 2x - 3$.

Answer. $\frac{5}{6}$; $2x + 6$; 1.

1.2 Parity and prime numbers

Key point · Definitions. An integer n is **even** if it can be written $2k$, with $k \in \mathbb{Z}$, and **odd** if it can be written $2k+1$. A **prime number** is a natural number greater than or equal to 2 that has exactly two positive divisors: 1 and itself.

The prime numbers less than 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Key point · Property to prove. The square of an odd integer is odd.

Proof to write.

Solution. If $n = 2k+1$, then $n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$. Since $2k^2+2k \in \mathbb{Z}$, n^2 is odd. The final form justifies the parity; a few numerical trials are not enough to prove the statement.

Method · Testing whether an integer is prime. For $n \geq 2$, it is enough to test the prime divisors less than or equal to $\sqrt{n}$. Indeed, if $n = ab$ with $1 < a \leq b$, then $a \leq \sqrt{n}$; a has a prime divisor which also divides n.

Example 2. Show that 157 is prime and that 161 is not.

Solution. Since $12^2 < 157 < 13^2$, we test $2, 3, 5, 7, 11$. None divides 157: for 7, $157 = 7 \times 22 + 3$; for 11, $157 = 11 \times 14 + 3$. So 157 is prime. On the other hand, $161 = 7 \times 23$ is composite.

1.3 GCD and irreducible fractions

Key point · Definition and convention. For two integers a, b **not both zero**, $\gcd(a,b)$ is the greatest of their common positive divisors. We have $\gcd(a,b) = \gcd(|a|,|b|)$ and $\gcd(a,0) = |a|$ if $a \neq 0$. By convention, we set $\gcd(0,0) = 0$; this last case does not come under the definition by the greatest divisor.

For example, $\gcd(18,24) = 6$, and changing the signs of 18 or 24 does not change this result.

Key point · Prime factorisation – accepted. Every natural number greater than or equal to 2 is a product of prime numbers. This factorisation is unique up to the order of the factors.

Example 3. Factorise 140, 150, 630 and 120 into prime factors.

Solution. $140 = 2^2 \times 5 \times 7$; $150 = 2 \times 3 \times 5^2$; $630 = 2 \times 3^2 \times 5 \times 7$; $120 = 2^3 \times 3 \times 5$. We divide successively by prime numbers and check the final product.

Method · Making a fraction irreducible. For $b \neq 0$, divide the numerator a and the denominator b by $\gcd(a,b)$. The fraction obtained has a numerator and a denominator that are coprime.

Example 4. Make $\dfrac{140}{630}$, $\dfrac{84}{126}$ and $\dfrac{-120}{150}$ irreducible.

Solution. The GCDs are 70, 42, 30 respectively. Thus $140/630 = 2/9$, $84/126 = 2/3$ and $-120/150 = -4/5$. For $140/630$, we can also cancel the factors $2, 5, 7$ in the factorisations.

1.4 Set vocabulary in context

Key point · Definitions and notation. $x \in A$ means that x is an element of A; $A \subseteq B$ means that every element of A belongs to B. $\varnothing$ is the empty set.

- $A \cap B$: elements belonging to A **and** to B;
- $A \cup B$: elements belonging to A **or** to B (possibly both);
- in a reference set E, $\overline{A} = E \setminus A$ contains the elements of E that are not in A;
- for a finite set, $\mathrm{Card}(A)$ is its number of elements.

Example 5. In $E = \{1,2,3,4,5,6\}$, let $A = \{2,4,6\}$ and $B = \{3,6\}$. Give $A \cap B$, $A \cup B$, $\overline{A}$ and $\mathrm{Card}(A)$. Are the statements $2 \in A$, $\{2\} \subseteq A$ and $A \subseteq B$ true?

Solution. $A \cap B = \{6\}$, $A \cup B = \{2,3,4,6\}$, $\overline{A} = \{1,3,5\}$, $\mathrm{Card}(A) = 3$. The first two statements are true. The third is false, for example $2 \in A$ and $2 \notin B$.

Key point · Pair and Cartesian product. A pair (a,b) is ordered: in general $(a,b) \neq (b,a)$. The Cartesian product $A \times B$ is the set of pairs (a,b) with $a \in A$ and $b \in B$. For finite sets, $\mathrm{Card}(A \times B) = \mathrm{Card}(A)\,\mathrm{Card}(B)$.

Example 6. Write all the elements of $\{1,2\} \times \{3,4,5\}$.

Solution. $\{(1,3),(1,4),(1,5),(2,3),(2,4),(2,5)\}$. There are $2 \times 3 = 6$ pairs.

1.5 Algorithms and putting it to use

Method · Python: integer, Boolean and condition. `a % b` is the remainder of the Euclidean division for integers and $b \neq 0$. A test returns a Boolean: `True` or `False`. Equality is tested with ==; assignment uses =.

```
def est_multiple(a, b):
    if b == 0:
        return a == 0
    return a % b == 0
```

Example 7. Predict the results of `est_multiple(35, 5)` and `est_multiple(35, 6)`, then of `est_multiple(0, 7)`, `est_multiple(7, 0)` and `est_multiple(0, 0)`. Explain the role of the condition.

Solution. The results are `True`, `False`, `True`, `False`, `True`. The case $b = 0$ avoids calculating a remainder modulo zero, which is impossible, and applies the definition: only 0 is a multiple of 0.

Example 8 · Problem. We have 84 notebooks and 126 pencils. We want to make as many identical packs as possible, using all the material. How many packs can we make? What does each contain?

Solution. The number of packs divides 84 and 126. Its maximum is therefore $\gcd(84, 126) = 42$. Each pack contains $84/42 = 2$ notebooks and $126/42 = 3$ pencils. The check $42 \times 2 = 84$, $42 \times 3 = 126$ confirms that all the material is used.

Teacher note. Diagnostic: have students state "divides" and "is a multiple of". Help: start by testing 2, 3, 6 packs. Extension: prove that $n(n+1)$ is even by distinguishing the two parities of n.